Diffusion Proposals for Redistricting Markov Chains

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Abstract

Recombination (ReCom) chains are the standard Markov chain Monte Carlo tool for sampling redistricting plans, and they are routinely run under a Metropolis filter that nominally targets a compactness-tempered distribution. We introduce a proposal built from graph diffusion. One convex energy on the merged pair of districts, a kernel-weighted Laplacian smoothness term plus a fidelity term that anchors each move to the current plan, is minimized under global population balance and rounded at random. The diffusion view yields a small theory of the proposal: its optimum is a single linear solve whenever balance is slack, the expected number of units a proposal moves is an explicit high-pass filter norm with a provable ceiling, and mobility is governed by feasibility rather than by the Metropolis ratio, which gives a tuning rule. Correcting the rounding with an exact Metropolis–Hastings step gives, to our knowledge, the first optimization-based redistricting proposal with a detailed-balance proof. Mixed with the forest-lifted recombination move Metropolized to the same target, it yields a reversible sampler for kernel-weighted compactness distributions, validated against enumerated targets. On Iowa it roughly doubles the effective samples per second of the forest move alone under perimeter and density targets, and adds nothing on the spanning-tree-weighted measure. The reason is structural, and we measure it: the tree move is an independence sampler from the untempered spanning-tree measure, so its acceptance decays exponentially with the size of the merged pair under any compactness-tempered target, while the local move has a dial for the same effect. On an enumerated test bed we also show, at the level of distributions, that plain and tempered ReCom do not sample their nominal target and sit close to the spanning-tree-weighted measure.

1 Introduction

1.1 The redistricting problem and ReCom's role

Political redistricting seeks a partition of a region into districts that are contiguous, population balanced, compact, and respectful of communities of interest. Computational approaches model the problem on the dual adjacency graph $G = (V, E)$ of geographic units. A valid k-district plan is a partition $P = \{V_1, \dots, V_k\}$ of V with each $G[V_i]$ connected and the populations $p(V_i)$ nearly equal; $\mathcal{P}_k(G)$ denotes the space of valid plans, and even four districts over the 99 counties of Iowa admit roughly 10^{24} of them [25]. Markov chain Monte Carlo is the standard tool for sampling it. Flip chains [24] mix slowly; Recombination (ReCom) [20] merges two adjacent districts, draws a uniform spanning tree of the merged subgraph and cuts a balanced edge, and is now the workhorse of the field, with reversible [10], forest-lifted [4] and multi-scale [3] extensions.

ReCom has two blind spots that this paper addresses. The first is that the uniform spanning-tree split treats every edge of the merged subgraph identically, so the proposal is determined by topology alone, while shared boundary length, population density and administrative boundaries all carry signal.

County-weighted trees [17], region-aware surcharges [36] and SpecReCom [19] inject some of it, but each shifts the chain's stationary distribution in an uncharacterized way. The second blind spot is shared by the exact versions of the tree move: they propose from one fixed law, the spanning-tree measure on balanced splits, and whenever the target is not close to that law the Metropolis step must undo the mismatch on every proposal. We show that the resulting acceptance decays exponentially in the size of the merged pair, which is the structural reason an exact sampler for compactness targets needs a second kind of move.

1.2 Contribution

We introduce a diffusion proposal for ReCom-style MCMC and the exact sampler built on it. The core object is one Laplacian energy on the merged-pair subgraph $H = G[V_i \cup V_j]$,

$$E(x) \;=\; \alpha\, x^\top L\, x \;+\; \beta\, \|x - x_0\|^2, \qquad (1-\varepsilon)\bar{p} \;\le\; p^\top x \;\le\; (1+\varepsilon)\bar{p}, \tag{1}$$

where L is the kernel-weighted Laplacian of H, x_0 encodes the current bipartition, and the balance constraint is anchored to the global per-district ideal. The Laplacian term counts (or weights) cut edges; the fidelity term keeps each move local. Four contributions follow.

1. **A proposal with a theory of its own behavior** (§4.3). The optimum of the energy is one implicit Euler step of the graph heat equation with timestep $\tau = \alpha/\beta$. It is a single linear solve whose output lies in the unit cube automatically (Lemma 4.1); under single-shot Bernoulli rounding the expected number of units moved is an explicit high-pass filter norm, growing like 2τ times the pair's boundary for small τ and bounded by half the pair for any τ (Proposition 4.2, Corollary 4.1); and the chain's mobility is a feasibility effect with an interior optimum in τ, given to within counting noise by a two-swap law (§4.6).

2. **An exactly reversible correction** (§4.5). Correcting the rounding with an exact Metropolis–Hastings step yields Chain E, to our knowledge the first optimization-based redistricting proposal with a detailed-balance proof (Proposition 4.3), verified to machine precision on an enumerated fiber and, by a batch-means test on a million-step run, indistinguishable from its target (§5.2).

3. **An exact sampler for kernel-weighted compactness targets, and a measurement of when its local component matters** (§4.7, §5.4, §5.5). We lift Chain E to the spanning-forest state space and mix it with the forest-lifted recombination move [4] Metropolized to an arbitrary kernel-weighted target. On Iowa, under a criterion fixed before the runs, the mixture gives 1.6 to 2.3 times the forest move's effective samples per second under perimeter and density targets on two seeds, and 0.8 to 0.9 times under the spanning-tree-weighted tempered measure. The reason is structural: the forest move is an independence sampler from the untempered spanning-tree measure, its acceptance on $N \times N$ grids falls like $e^{-0.24N}$ under compactness-tempered targets with or without tree weights, and the per-proposal mismatch in its Hastings ratio grows with the boundary length. The local move loses acceptance with pair size too, but has a dial for it, the move-size law; the tree move has none.

4. **A ground-truth calibration of what the ReCom family samples** (§5.1, §5.6). On a fully enumerated 5×5 fiber the tempered chain's cut distribution is 220 noise-floor standard deviations from its nominal target and close to the tree-weighted tempered measure, while the exact chain is indistinguishable from the nominal target; on Iowa the exact sampler puts the nominal target's mean cut count near 88 and the tree-weighted measure's at 41, where tempered ReCom sits at 40.

Positioning. A deterministic optimization that sets the parameters of a random proposal which is then corrected exactly has ancestors outside redistricting: randomize-then-optimize [5] and Zanella's informed proposals [48], of which Chain E is an instance on the space of plans, so our priority claim is scoped to redistricting. Within redistricting, the known-target samplers (Metropolized forest recombination, RevReCom, the cycle walk, the marked edge walk, BUD [4, 10, 21, 34, 1]) achieve computable transition probabilities by lifting to spanning-forest state spaces, while Chain E does so at the partition level; the two routes are complementary and the forest move is the global component of our sampler. Bonsai sampling [18] draws independent plans from a specified law without a chain.

Roadmap. §2 reviews related work and §3 sets up notation. §4 develops the proposal, its theory, the exact chain, and the sampler; §5 reports the calibration, the convergence validation, the kill test against the forest move, the scaling experiment, and the Iowa calibration. Appendices hold hyperparameters and a reproducibility checklist.

2 Related work

MCMC for redistricting. Flip chains [24] mix poorly [37]; ReCom [20] is standard, with multi-scale [3], forest-lifted [4], sequential Monte Carlo [32, 38, 9], parallel-tempering [14] and reversible [10] variants. The compactness-tempered target practitioners run under a Metropolis filter originates in Herschlag et al. [29]; the 50-State Atlas [33] is the standard benchmark. The 2022–2026 literature's theme is that a sampler is only as defensible as its target, and RevReCom, forest recombination, the cycle walk [21] and the marked edge walk [34] all pay for computable transition probabilities with lifted state spaces. Cannon, Pegden and Tucker-Foltz [12] give a polynomial-time sampler for the spanning-tree measure on grids with constant k, Gold [27] exhibits configurations that trap recombination for exponentially many steps, Akitaya et al. [2] prove reachability PSPACE-complete when k and the step size grow with the graph, and mixing bounds for the chains used in practice remain open.

Weighted and geography-aware proposals. County-weighted trees [17], GerryChain's region-aware surcharges [36] and Tapp's compactness-aware split [42] weight edges inside the ReCom proposal; in all three the induced stationary distribution is uncharacterized. What the unweighted tree proposal induces is better understood: Clelland et al. [16] measured ReCom's sampling probability to decay roughly exponentially in cut length, Procaccia and Tucker-Foltz [40] proved that compact plans have many spanning trees, and Tapp, Proebsting and Ramsay [43] study how tolerance and county settings move ensembles. Our calibration (§5.1) makes the first observation exact on an enumerated fiber, and our scaling experiment (§5.5) turns the second into a statement about acceptance: the exponential that makes compact plans tree-rich is what an exact tree move pays, per proposal, to target anything else. The known-target samplers above handle only tree-count-based families; the gap this paper fills is a proposal that takes an arbitrary nonnegative edge kernel as input together with an exact accept/reject step against a stated target.

Optimization and spectral lines. An operations-research literature treats districting as exact optimization [28, 41, 46, 45, 22], and hybrids with sampling are emerging: short bursts [11] use ReCom as an optimizer, and Brous and Shmoys [7] put a Gurobi integer program inside a local search. The integer form of our energy belongs to that matheuristic line (§6.4); its proximal term is the quadratic analogue of local branching [26]. On the spectral side, SpecReCom [19] replaces the tree with a Fiedler-vector cut of a randomly weighted Laplacian, and diffuse-interface segmentation [13, 6, 35, 39] supplies the Laplacian-plus-fidelity energy we use; Jacobs and Walch [31, 30] brought threshold dynamics to redistricting as a

continuum PDE method outside MCMC. To our knowledge this is the first use of the graph Ginzburg–Landau lineage as a proposal inside redistricting MCMC. The cut-edge count as a compactness measure is due to Duchin and Tenner [23].

3 Background: plans and ReCom

3.1 Dual graph and k-district plans

Let $G = (V, E)$ be the dual graph of a state, with vertices representing geographic units (precincts, census blocks, counties) and edges encoding shared boundaries. Each vertex v carries a population $p_v \geq 0$ and may carry additional attributes (demographic vectors, shapefile geometry). A *k-district plan* is a partition $P = \{V_1, \ldots, V_k\}$ of V such that each induced subgraph $G[V_i]$ is connected (*contiguity*) and the populations satisfy

$$(1-\varepsilon)\bar{p} \;\leq\; p(V_i) \;\leq\; (1+\varepsilon)\bar{p}, \qquad \bar{p} := \frac{p(V)}{k}, \tag{2}$$

for a tolerance $\varepsilon \in [0, 1)$. We write $\mathcal{P}_k(G; \varepsilon)$ for the set of valid plans. The *cut edges* of a plan are

$$\partial P \;=\; \{(u, v) \in E : P(u) \neq P(v)\},$$

and we call $|\partial P|$ the *cut-edge count* of P. This quantity is the discrete analog of the perimeter of the district boundaries and serves as our primary compactness metric [23]. For a nonnegative edge kernel w we write $w(\partial P) = \sum_{e \in \partial P} w_e$ for the *weighted cut*.

3.2 Recombination (ReCom)

Recombination [20] is a Markov chain on $\mathcal{P}_k(G; \varepsilon)$. One step selects two adjacent districts V_i, V_j, forms $H = G[V_i \cup V_j]$, draws a uniform spanning tree of H by Wilson's algorithm [47], searches its edges for one whose removal leaves two ε-balanced components (redrawing the tree if none exists), and replaces V_i, V_j by the two components. Many units move at once and contiguity is automatic, which is the advantage over flip chains.

Chains compared. Table 1 lists the chains of this paper.

Chain	Method	Mechanism	Stationarity claim
V	Vanilla ReCom [20]	spanning-tree split, always accept	unknown target
A	Tempered ReCom	ReCom + MH on cuts (symmetric-q)	approximate
E	QP-MH (this work, §4.5)	QP + Bernoulli rounding + exact MH	**exact**
T	Partition-level tree move (§4.7.1)	uniform tree, uniform edge, tree-count MH	**exact**
F	Forest move (§4.7.2)	forest-lifted ReCom, balanced edge, exact MH	**exact**
H	Hybrid (this work, §4.7)	mixture of F and lifted E, any kernel-weighted target	**exact**

Table 1: The chains compared. Chains E, T, F and H carry exact Metropolis–Hastings corrections; Chain A relies on the symmetric-q approximation (§4.5) and Chain V has no stated target. Chain T explains Chain A; Chain F is the baseline the sampler is judged against.

Tempered ReCom uses a Metropolis–Hastings filter targeting $\pi(P) \propto \exp(-\lambda|\partial P|)$ with the symmetric-q approximation, which treats forward and reverse proposal probabilities as equal, at the locked-in $\lambda = 0.10$ of Appendix A. All chains enforce the global two-sided balance window of Equation (4) on every move; we note (§5) that the raw GerryChain `recom` proposal alone does not.

4 The QP-diffusion model, its theory, and the sampler

4.1 Locality kernels

Choosing the edge weights w_{ij} is the most important modeling decision. The kernel determines what *smoothness* means and therefore controls how district boundaries move under the relaxation. We restrict all weights to geographic neighbors,

$$w_{ij} = 0 \quad \text{if } (v_i, v_j) \notin E, \tag{3}$$

ensuring that diffusion operates only through geographic adjacency and preventing long-range interactions that could produce disconnected districts.

Uniform kernel (k_{unif}). $w_{ij} = 1$ for every edge. The Laplacian L is the combinatorial graph Laplacian; $x^\top Lx$ is exactly the cut edge count. In the $\beta \to 0^+$ limit the proposal approaches a balanced Fiedler-style split.

Perimeter kernel (k_{perim}). $w_{ij} = |\partial v_i \cap \partial v_j|$ (shared-boundary length). The Laplacian quadratic form $x^\top Lx$ is then the *geographic perimeter* of the partition, weighted by boundary segment length. This kernel discourages fragmented districts and favors boundaries that cut along short segments.

Density kernel (k_{dens}). Let ρ_v denote the population density at vertex v (population per unit area). A Gaussian density kernel

$$w_{ij} \;=\; |\partial v_i \cap \partial v_j| \cdot \exp\Big(-\frac{(\rho_i - \rho_j)^2}{\sigma^2}\Big)$$

encourages cuts along sharp density gradients (urban–rural boundaries) while preserving the topological role of shared boundary length.

Demographic kernel (k_{demog}). A Gaussian similarity kernel on a vector of attributes d_v encourages communities of interest. We define it for completeness and do not use it: a kernel built on race would make race the predominant districting criterion, which the *Shaw v. Reno* line of cases subjects to strict scrutiny, so any such use needs legal review and is more defensible for attributes that are legitimate factors in the jurisdiction. Nonnegative linear combinations of kernels remain valid.

4.2 The convex QP

Let $H = G[V_i \cup V_j]$ be the merged subgraph of two adjacent districts. Let L_H be the kernel-weighted Laplacian of H, let $x_0 \in \{0,1\}^{|V_H|}$ encode the current bipartition (1 for vertices in V_i, 0 for vertices in V_j), and let p be the population vector restricted to H. Fix weights $\alpha, \beta > 0$ and balance tolerance $\varepsilon \in (0,1)$. Crucially, the balance constraint is anchored to the *global* per-district ideal $\bar{p} = p(V)/k$ from Equation (2), not to $p(V_H)/2$. Both sides of the split must individually satisfy the plan-validity bound, which yields the two-sided window

$$\ell \;=\; \max\big((1-\varepsilon)\bar{p},\; p(V_H) - (1+\varepsilon)\bar{p}\big), \qquad u \;=\; \min\big((1+\varepsilon)\bar{p},\; p(V_H) - (1-\varepsilon)\bar{p}\big). \tag{4}$$

Anchoring to $p(V_H)/2$ instead (a tempting simplification, and one present in an earlier version of our own implementation) only bounds the local split. Because $p(V_H)$ itself drifts as the chain moves population between pairs, district populations can ratchet outside the global tolerance by a factor of up to $(1+\varepsilon)$ per move on heterogeneous graphs. Our experiments audit the global bound at every step (§5).

Definition 4.1 (Continuous QP-diffusion)**.** *The continuous QP-diffusion bipartition of H is*

$$x^\star \;=\; \arg\min_{x\in[0,1]^{|V_H|}} \;\alpha\, x^\top L_H\, x \;+\; \beta\,\|x - x_0\|_2^2 \tag{5}$$

$$\text{subject to } \;\ell \;\le\; p^\top x \;\le\; u, \tag{6}$$

with ℓ, u as in Equation (4).

The first term in (5) is the kernel-weighted Laplacian smoothness energy. For binary x it equals the weighted cut count $\sum_{(u,v)\in E(H)} w_{uv}\,\mathbf{1}[x_u \neq x_v]$, and the continuous relaxation $x \in [0,1]^{|V_H|}$ extends this to a smooth objective on the unit cube. The second term is the *fidelity term*, which penalizes deviation from the current bipartition and ensures that each proposed move is local. The balance constraint enforces ε-balance on the proposed split.

Proposition 4.1 (Convexity and well-posedness)**.** *The objective in (5) is strictly convex on $[0,1]^{|V_H|}$ (since $\beta > 0$ and L_H is positive semidefinite). The feasible region defined by (6) is a nonempty polytope whenever the current plan is valid (it then contains x_0). Hence the QP admits a unique minimizer $x^\star$, computable in polynomial time by any convex QP solver.*

Spectral interpretation. Projecting the interior optimality condition $(\alpha L_H + \beta I)x^\star = \beta x_0$ onto the eigenbasis $\{\phi_k\}$ of L_H with eigenvalues $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$ yields the closed-form frequency response

$$\hat{x}_k^\star \;=\; \frac{\beta}{\alpha\lambda_k + \beta}\,\hat{x}_{0,k}, \qquad \hat{y}_k := \phi_k^\top y. \tag{7}$$

The QP therefore acts as a low-pass spectral filter on the prior x_0, with cutoff controlled by the ratio α/β. Equivalently, with $\tau := \alpha/\beta$, the optimum satisfies $(x^\star - x_0)/\tau = -L_H x^\star$, the implicit Euler discretization of the graph heat equation $\dot{x} = -L_H x$. Inspecting the leading eigenvectors is a sanity check on the kernel: noise rather than coherent regions means an uninformative kernel.

4.3 Theory of the diffusion proposal

The spectral identity is usually read as interpretation. Worked through, it predicts how the proposal behaves, and every prediction below is checked against the solver. Write $\tau := \alpha/\beta$ for the diffusion timestep, $L = L_H$ for the kernel-weighted Laplacian of the merged pair, $n = |V_H|$, and $\mathbf{1}_i \in \{0,1\}^n$ for the indicator of V_i, so that $x_0 = \mathbf{1}_i$.

Lemma 4.1 (The box constraints never bind)**.** *For any nonnegative kernel and any $\tau > 0$, the matrix $(I + \tau L)^{-1}$ is symmetric, entrywise nonnegative, and has unit row sums. Hence the unconstrained minimizer $x^\star = (I + \tau L)^{-1} x_0$ of (5) is a convex combination of the entries of x_0 and lies in $[0,1]^n$, and the QP of Definition 4.1 reduces to this single linear solve whenever the balance constraint (6) is slack at $x^\star$.*

Proof. $I + \tau L$ is symmetric with positive diagonal, nonpositive off-diagonal entries, and row sums equal to one, so it is strictly diagonally dominant, hence a nonsingular M-matrix, and its inverse is entrywise nonnegative. Since $L\mathbf{1} = 0$ we have $(I + \tau L)\mathbf{1} = \mathbf{1}$ and so $(I + \tau L)^{-1}\mathbf{1} = \mathbf{1}$. Each row of the inverse is therefore a probability vector and $x_v^\star$ is a weighted average of entries of $x_0 \in \{0,1\}^n$. The objective is strictly convex, so if the unconstrained minimizer satisfies the one remaining constraint it is the constrained minimizer. □

Proposition 4.2 (Expected move size)**.** *Under the single-shot Bernoulli rounding of §4.5, and whenever the balance constraint is slack, the expected number of units of H that a proposal reassigns is*

$$\mathbb{E}[\#\text{flips}] \;=\; \|x^\star - x_0\|_1 \;=\; 2\,\mathbf{1}_i^\top (I+\tau L)^{-1}\mathbf{1}_j \;=\; \frac{2\,|V_i|\,|V_j|}{n} \;-\; 2\sum_{k\geq 2}\frac{(\phi_k^\top \mathbf{1}_i)^2}{1+\tau\lambda_k}, \tag{8}$$

where (λ_k, ϕ_k) are the eigenpairs of L with $\lambda_1 = 0$.

Proof. A unit with $x_{0,v} = 0$ flips with probability $x_v^\star$ and one with $x_{0,v} = 1$ with probability $1 - x_v^\star$; both equal $|x_v^\star - x_{0,v}|$ because $x_v^\star \in [0,1]$ by Lemma 4.1, which gives the first equality. Write $P = (I+\tau L)^{-1}$. For $v \in V_i$, $1 - x_v^\star = \sum_{u\in V_j} P_{vu}$ by unit row sums, and for $v \in V_j$, $x_v^\star = \sum_{u\in V_i} P_{vu}$; summing and using symmetry of P gives $2\,\mathbf{1}_i^\top P\,\mathbf{1}_j$. Expanding $P = \sum_k \phi_k\phi_k^\top/(1+\tau\lambda_k)$ with $\phi_1 = \mathbf{1}/\sqrt{n}$ gives the $k=1$ term $|V_i||V_j|/n$, and since $\phi_k \perp \mathbf{1}$ for $k \geq 2$, $\phi_k^\top \mathbf{1}_j = -\phi_k^\top \mathbf{1}_i$, which gives the sum. Clipping at $\delta = 10^{-12}$ changes the expectation by at most δn. □

Corollary 4.1 (Small-timestep law, monotonicity, and ceiling)**.** *Let $|\partial_w(V_i, V_j)|$ be the boundary of the pair measured in the proposal's kernel. Then $\mathbb{E}[\#\text{flips}]$ is increasing in τ,*

$$\mathbb{E}[\#\text{flips}] \;=\; 2\tau\,|\partial_w(V_i,V_j)| + O(\tau^2) \quad (\tau \to 0), \qquad \mathbb{E}[\#\text{flips}] \;\leq\; \frac{2\,|V_i|\,|V_j|}{n} \quad \textit{for every } \tau,$$

with equality in the limit $\tau \to \infty$. For a balanced pair the ceiling is $n/2$, half the merged pair.

Proof. Each subtracted term in (8) is nonnegative and decreasing in τ, which gives monotonicity and the bound, and the terms vanish as $\tau \to \infty$. For the small-τ law, expand $(1+\tau\lambda_k)^{-1} = 1 - \tau\lambda_k + O(\tau^2)$; the τ^0 terms cancel because $\sum_{k\geq 2}(\phi_k^\top \mathbf{1}_i)^2 = |V_i| - |V_i|^2/n = |V_i||V_j|/n$, and the τ^1 term is $2\tau\sum_k \lambda_k(\phi_k^\top \mathbf{1}_i)^2 = 2\tau\,\mathbf{1}_i^\top L\,\mathbf{1}_i = 2\tau|\partial_w|$. □

Three consequences follow. The continuous proposal is far cheaper than "solve a QP" suggests, since it is one linear solve in the common case. The move size is a design dial with a known scale, so to propose moves of about m units one sets $\alpha/\beta \approx m/(2|\partial_w|)$ while the small-timestep law holds; §5.5 uses this dial. And the dial has a hard stop. No choice of α and β can lift the expected move above half the merged pair, so mobility cannot be bought by tuning, which is the structural reason a global move is needed alongside the exact local chain (§4.6, §4.7). The kernel enters through $|\partial_w|$, so a perimeter kernel takes larger steps exactly where the geographic border is long.

Verification. Across 10 766 linear solves the unconstrained optimum never left the unit cube; on 120 pairs with balance slack, Gurobi's solution and the linear solve agree to 1.1×10^{-4}. The small-timestep law's median relative error is 2.5% at $\tau = 0.01$, 7.4% at 0.03 and 24% at 0.1 (Figure 1). At the locked setting the expected move is 0.42 units on the 5×5 fiber and about one unit on Iowa, which is the quantitative content of Chain E's stickiness.

4.4 Deterministic rounding, and why it blocks the correction

The continuous optimum $x^\star \in [0,1]^{|V_H|}$ must be rounded to a bipartition. The natural deterministic choice is a Cheeger sweep [15]: sort the entries, scan the thresholds, keep the balanced connected candidate with the smallest cut. As a proposal it has a defect no repair fixes: an argmin-plus-sweep map is deterministic, so the reverse proposal has zero probability and the Hastings ratio degenerates. A chain built on it is a heuristic with no stationarity claim. The same defect is why the forest move of §4.7.2 keeps the tree draw and the cut-edge draw random rather than optimized.

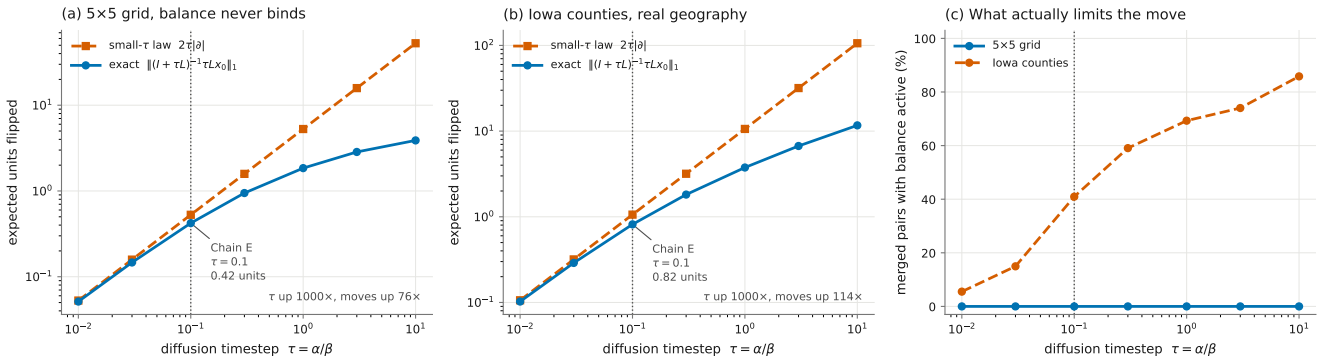


Figure 1: The move-size law against the solver. (a) On the enumerated 5×5 fiber, 1 411 merged pairs drawn from 200 plans; the exact filter norm of (8) and the small-timestep law $2\tau|\partial_w|$ agree for $\tau \leq 0.03$ and separate as the gain saturates. Raising τ by $1000\times$ raises the move by $76\times$, and the last decade buys $2.1\times$. The locked Chain E setting $\tau = 0.1$ is marked. (b) The same on Iowa, 127 pairs from 25 tree-partition plans. (c) Fraction of merged pairs whose balance constraint is active. On the uniform grid it never is, so the proposal is a linear solve at every step; on real county geography it is active for 41% of pairs at $\tau = 0.1$ and 86% at $\tau = 10$, so there the population constraint, not the diffusion, caps the move.

4.5 The exactly reversible QP-MH chain (Chain E)

Replacing the deterministic rounding of §4.4 with *single-shot randomized rounding* restores a tractable, everywhere-positive proposal density and yields an exactly reversible chain. One step of Chain E is as follows.

1. Pick an adjacent district pair (V_i, V_j) uniformly at random from the $|\mathrm{bp}(P)|$ boundary pairs of the current plan P;
2. Solve the (deterministic) QP of Definition 4.1 on H for $x^\star \in [0,1]^{|V_H|}$;
3. Draw the proposal $x' \in \{0,1\}^{|V_H|}$ by independent Bernoulli rounding from the *clipped* optimum, $x'_v \sim \mathrm{Bern}(\mathrm{clip}(x^\star_v; \delta, 1-\delta))$ with $\delta = 10^{-12}$, *once*, with no retry on infeasibility. Clipping is part of the proposal: the optimum can have coordinates exactly 0 or 1, and sampling from the same clipped vector whose density enters the acceptance ratio gives every binary x' positive probability;
4. Accept with probability
$$\min\Big(1,\ \frac{\pi(P')}{\pi(P)}\,\frac{q(x_0 \mid P', \{i,j\})}{q(x' \mid P, \{i,j\})}\,\frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big), \tag{9}$$
where $q(\cdot \mid P, \{i,j\})$ is the Bernoulli product density induced by the QP optimum at the corresponding state, and proposals that violate contiguity or the balance window (4) are rejected outright.

Here π is any target on plans; the paper uses $\pi_w(P) \propto \exp(-\lambda\, w(\partial P))$ for a kernel w, which reduces to the tempered cut-count target for the uniform kernel, and in §4.7 also the tree-weighted variant.

Proposition 4.3 (Detailed balance of Chain E)**.** *Chain E satisfies detailed balance with respect to any target π on $\mathcal{P}_k(G;\varepsilon)$ that is positive on valid plans.*

Proof. Fix the deterministic kernel w and write Ω for the space of valid plans, taken as *labeled* assignments so that a Bernoulli draw and its complement are distinct states and the proposal map is a bijection; π is label-invariant, so forgetting labels after sampling recovers the unlabeled law. For distinct $P, P' \in \Omega$,

a step changes only the two selected districts, so if P and P' differ on more than two districts, or on two whose union H is disconnected and hence never selected, then $K(P,P') = K(P',P) = 0$. Otherwise the differing pair $\{i,j\}$ is the unique pair that can generate either move, and it is adjacent in both plans because every bipartition of the connected graph H into nonempty parts is crossed by an edge. Three facts make the transition probability explicit. First, the feasible set $\{x \in [0,1]^{V_H} : \ell \le p^\top x \le u\}$ of the QP depends on $(H, \varepsilon, \bar{p})$ only, not on the endpoint, and contains both $x_0(P)$ and $x_0(P')$ because the global window (4) is exactly the validity condition of the two sides. This is where globality is essential: a window anchored to the merged-pair half $(1 \pm \varepsilon)\, p(V_H)/2$ lets a valid plan fail its own pair's window, zeroing a reverse density whose forward counterpart is positive. Second, the objective has Hessian $2(\alpha L_H + \beta I) \succ 0$, so the optimum is unique and a deterministic function of the endpoint and the pair. Third, clipping before sampling puts every Bernoulli factor in $[\delta, 1-\delta]$, so every binary labeling of V_H has probability at least $\delta^{|V_H|} > 0$, and because the rounding is single-shot the density $q_\to = q(x_0(P') \mid P, \{i,j\})$ is the exact probability that the proposal from P equals P': no feasibility renormalization, whose state-dependent constant would not cancel, enters. Hence

$$\pi(P)\, K(P,P') \;=\; \frac{\pi(P)\, q_\to}{|\mathrm{bp}(P)|}\, \min\Big(1,\; \frac{\pi(P')\, q_\leftarrow\, |\mathrm{bp}(P)|}{\pi(P)\, q_\to\, |\mathrm{bp}(P')|}\Big) \;=\; \min\Big(\frac{\pi(P)\, q_\to}{|\mathrm{bp}(P)|},\; \frac{\pi(P')\, q_\leftarrow}{|\mathrm{bp}(P')|}\Big),$$

which is symmetric in $(P, q_\to) \leftrightarrow (P', q_\leftarrow)$. Draws that leave a side empty, disconnected or outside the window land on states of zero target mass and are rejected with probability one, contributing only to the diagonal; a solver failure suppresses the forward and reverse moves identically because the solve is the same deterministic function at both endpoints. Detailed balance and stationarity follow [44]. □

Remark 4.1 (Verification and ergodicity)**.** *The proof treats the solve as exact; the implementation is deterministic (fixed kernel, no jitter, sorted pair enumeration), which is what the argument uses. We assembled the full transition kernel of the implementation on the enumerated* 2×3 *grid (*$k = 2$*,* $\varepsilon = 1/3$*, 18 labeled valid states) and verified* $\pi(P)K(P,P') = \pi(P')K(P',P)$ *entrywise to a maximum relative violation of* 1.5×10^{-15}*. Detailed balance gives stationarity, not ergodicity: within a pair every relabeling has positive probability, but irreducibility of Chain E across pairs on general* (G, k, ε) *is not established, and the principled route to global mobility is the mixture of §4.7.*

Chain E trades mobility for exactness: a proposal must be feasible in a single Bernoulli draw and Corollary 4.1 caps its reach, so the chain is a strictly local sampler (§4.6), and we present it as the local component. Mixing it with a global kernel preserves π only if that kernel is also π-invariant; RevReCom is reversible for its spanning-tree measure, not for an arbitrary π, so the construction of §4.7 Metropolizes the tree move toward π using its computable Hastings ratio [10, 4].

On the tempered baseline (Chain A). Tempered ReCom applies $\min(1, \exp(-\lambda\Delta|\partial P|))$ over the ReCom proposal, which assumes a symmetric proposal. ReCom's is not: the true Hastings ratio involves spanning-tree counts of the affected districts [4, 10], so Chain A targets π only approximately; §5.1 quantifies the bias, and we keep the baseline because it is what practitioners run.

4.6 Mobility of the exact chain: feasibility, not acceptance

Proposition 4.2 says how far a proposal reaches. Mobility is that reach times the probability the proposal is kept, and the second factor has two causes of different kinds. A Bernoulli draw can produce a split that is disconnected or outside the balance window, in which case it is rejected with probability one before any coin is flipped. Or it can be a valid plan that loses the Metropolis coin. We measured both on the enumerated 5×5 fiber, with $\alpha = \tau$ and $\beta = 1$, over 3 000 steps per timestep from the same initial plan (Table 2).

τ	$\mathbb{E}$[#flips]	single-flip share	feasible | single	Pr[feasible]	accept | feasible	units moved / step	states
0.01	0.036	0.035	0.000	0.964	1.000	0.0000	1
0.03	0.126	0.110	0.000	0.884	1.000	0.0033	6
0.10	0.385	0.268	0.000	0.683	0.999	0.0093	12
0.30	0.909	0.349	0.000	0.437	0.989	**0.0353**	**38**
1.00	1.914	0.254	0.000	0.193	0.983	0.0313	37
3.00	2.672	0.168	0.000	0.074	0.986	0.0187	19

Table 2: Chain E against the diffusion timestep on the 5×5 fiber ($k = 5$, exact balance). "Feasible | single" is the feasibility rate of single-unit proposals, "accept | feasible" the Metropolis acceptance rate among feasible proposals, and "units moved / step" the product of acceptance and accepted move size, the quantity a tuning rule should maximize. "States" counts distinct fiber states reached in 3 000 steps. The locked setting of Appendix A is $\tau = 0.10$.

Three facts follow. First, every single-unit proposal is infeasible at every timestep, exactly as exact balance requires, since reassigning one cell produces districts of sizes 4 and 6. The leading-order move of the diffusion proposal is therefore always rejected on this fiber, and the chain moves only through second-order coordinated draws. This is the mechanism behind the coverage figures of §5.1, derived rather than observed. Second, the Metropolis ratio is never the bottleneck. Given a feasible proposal, acceptance is between 98 and 100 percent throughout. The chain's stickiness is a feasibility effect, and the natural phrase "large moves are rarely accepted" is wrong in an instructive way. They are rarely feasible, and when feasible they are accepted. Third, acceptance rate is a misleading diagnostic for this chain. At $\tau = 0.01$ it is 96 percent while the chain never leaves its initial state, because 96 percent of proposals flip nothing. We report units moved per step instead.

Units moved per step has an interior maximum near $\tau = 0.3$. The setting locked in Appendix A, $\tau = 0.1$, sits about $3.8\times$ below it, so on this fiber the chain can be improved for free; §5.3 shows the optimum does not transfer to Iowa, where the balance window binds more often at larger τ, and §5.5 shows that on large pairs the dial has to be turned the other way. Even at the optimum a thousand steps move about 35 units, and Corollary 4.1 says no timestep can lift the per-proposal reach above half the merged pair. Both statements matter. Chain E can be several times more mobile than shown, and several times is not enough, which is why a global move is required (§4.7). Under a loose tolerance single flips become feasible and mobility gains a first-order term in τ, which is why the same chain climbs steadily on Iowa (§5.6) while freezing on the exact-balance grid.

The mobility law. On an exact-balance fiber the cheapest feasible draw is a swap, one unit $u \in V_i$ and one unit $v \in V_j$ changing sides together, which preserves both populations. Rounding flips units independently with probabilities $p_v = |x^\star_v - x_{0,v}|$, so the probability that exactly $\{u, v\}$ flips is $p_u p_v \prod_{w \neq u,v}(1 - p_w)$, and the two-swap channel contributes

$$M_2 \;=\; 2 \sum_{\substack{u \in V_i,\, v \in V_j \\ \text{swap feasible}}} p_u\, p_v \prod_{w \neq u,v} (1 - p_w)\; \alpha(u, v) \tag{10}$$

units per step, where the sum runs over cross-boundary pairs whose swap keeps both districts connected and $\alpha(u, v)$ is the Metropolis probability of that swap, each of which costs one reverse linear solve. For small τ, $p_v = \tau b_v + O(\tau^2)$ with b_v the kernel weight of v's cross-boundary edges, so $M_2 = 2\tau^2 S_2 + O(\tau^3)$ with $S_2 = \sum_{\text{feasible swaps}} b_u b_v$, a coefficient computable from the current plan. We checked the law by running Chain E for 20 000 steps per timestep and evaluating (10) at every step for the pair chosen and the plan visited: measured units per step 0.0021, 0.0105 and 0.0320 at $\tau = 0.03$, 0.1 and 0.3 against $M_2 = 0.0018$, 0.0105 and 0.0326.

The two-swap channel accounts for the mobility at every timestep, four-unit draws contribute nothing measurable even at $\tau = 0.3$, and without the Metropolis factor the channel overpredicts by about 20%. The small-timestep form $2\tau^2 S_2$ holds only for $\tau \lesssim 0.03$. Under a loose tolerance single flips become feasible and the law gains a first-order channel.

4.7 Exact tree moves, the lifted local move, and the reversible hybrid

Corollary 4.1 says the local chain cannot be made mobile by tuning, so a global move is needed, and §4.5 says that move must be reversible for the same target. This subsection gives two exact tree moves, the one we use and the one that explains Chain A, lifts Chain E to the state space of the first, and states the sampler. It ends with the fact that organizes the experiments: an exact tree move proposes from one fixed law, and pays for every target that differs from it.

4.7.1 The partition-level tree move (Chain T)

Pick an adjacent pair $\{i, j\}$ uniformly from $\mathrm{bp}(P)$, draw a uniform spanning tree T of $H = G[V_i \cup V_j]$ by Wilson's algorithm, and pick one edge of T uniformly among *all* $|V_H| - 1$ tree edges. If the two components A, B both lie in the window (4), assign the labels i, j to them by a fair coin and propose; otherwise reject. Every spanning tree of H that yields $\{A, B\}$ on removal of one edge is a spanning tree of $G[A]$, a spanning tree of $G[B]$, and one crossing edge, so

$$q(P \to P') \;=\; \frac{\tau(A)\,\tau(B)\,|\partial(A,B)|}{2\,|\mathrm{bp}(P)|\,\tau(H)\,(|V_H|-1)},$$

where $\tau(\cdot)$ is the spanning-tree count by the Matrix–Tree theorem and $|\partial(A,B)|$ the number of edges between A and B. The reverse move shares H, so $\tau(H)$ and $|V_H| - 1$ cancel and the move is accepted with probability

$$\min\Big(1,\; \frac{\pi(P')}{\pi(P)}\;\frac{\tau(V_i)\,\tau(V_j)\,|\partial(V_i,V_j)|}{\tau(A)\,\tau(B)\,|\partial(A,B)|}\;\frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big),$$

computable from four determinants and two edge counts; the fair coin makes the move a bijection on labeled plans. The cut edge is drawn among all tree edges rather than among balanced ones, which is the point: GerryChain's `recom` draws among balanced edges, which multiplies the density by a tree-dependent normalization that does not cancel at the partition level, and that is why Chain A is only approximately correct. The price is acceptance, about 7% on the exact-balance 5×5 fiber and 1% on Iowa, so this move is not what practice runs and is kept here as a reference.

4.7.2 The forest move (Chain F)

Recovering balanced-edge selection with an exact density means lifting the state to a spanning forest, as Metropolized forest recombination does [4]. A labeled forest is $\mathcal{F} = (P, \{T_d\})$, where T_d is a spanning tree of $G[V_d]$ for every district. The fiber over a plan P has $\tau(P) = \prod_d \tau(V_d)$ forests, and the lifted target

$$\tilde{\pi}(\mathcal{F}) \;=\; \frac{\pi(P(\mathcal{F}))}{\tau(P(\mathcal{F}))} \tag{11}$$

has marginal π on plans. Every statistic we report is a function of the plan, so traces are comparable with the partition-level chains.

One step. Pick $\{i, j\}$ uniformly from $\mathrm{bp}(P)$ and draw a uniform spanning tree T of H. Let $B(T)$ be the set of edges of T whose removal leaves both components inside the window (4). If $B(T)$ is empty, draw again, up to M times, then reject. Otherwise pick $e \in B(T)$ uniformly, let $\{A, B\}$ be the components,

and assign the labels by a fair coin. A forest $\mathcal{F}' = (\ldots, T_A, T_B, \ldots)$ is reached from exactly those draws $(\hat{T}, \hat{e})$ with $\hat{T} = T_A \cup T_B \cup f$ for a crossing edge $f \in \partial(A, B)$ and $\hat{e} = f$, so

$$q(\mathcal{F} \to \mathcal{F}') \;=\; \frac{c_H}{2\,|\mathrm{bp}(P)|\,\tau(H)}\, S(T_A, T_B), \qquad S(T_A, T_B) \;=\; \sum_{f \in \partial(A,B)} \frac{1}{|B(T_A \cup T_B \cup f)|}, \tag{12}$$

where $c_H = (1-(1-p_H)^M)/p_H$ accounts for the retry, p_H being the probability that a uniform spanning tree of H has a balanced edge. The retry is exact because p_H depends on H and the window only, never on the endpoint, and the reverse move shares H, so c_H and $\tau(H)$ cancel. (This is not true of the local move's rounding, whose feasibility event depends on the QP optimum at the endpoint; that is why Chain E must be single-shot.) The Hastings ratio is

$$\frac{\pi(P')}{\pi(P)} \cdot \frac{\tau(V_i)\,\tau(V_j)}{\tau(A)\,\tau(B)} \cdot \frac{S(T_i, T_j)}{S(T_A, T_B)} \cdot \frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}, \tag{13}$$

where the tree-count ratio is the lifted target's, not the proposal's. Each S is a sum over the crossing edges of an $O(n)$ subtree-population pass, and the two determinants of the current districts are cached, so a step costs two determinants plus $O(|\partial|\,n)$. Our implementation differs from Autry et al. in one convention: the adjacent pair is drawn uniformly from boundary pairs, as in every other move of this paper, rather than by a uniform cut edge; both are exact and the ratio carries the corresponding factor.

Two targets. Besides π_w we also use the tree-weighted version $\pi_w^{\mathrm{tree}}(P) \propto \tau(P)\, e^{-\lambda\, w(\partial P)}$, the measure that §5.1 identifies as close to what tempered ReCom samples. Under π_w^{tree} the tree-count ratio in (13) cancels against the same factor in $\pi(P')/\pi(P)$. Both targets are handled by the same code path; the local move's acceptance under the tree-weighted target acquires the tree-count ratio instead, since it changes the plan and hence $\tau(P)$.

4.7.3 The lifted local move and the hybrid (Chain H)

Run Chain E on the plan against the chosen target. If it accepts, redraw the trees of the two changed districts uniformly by Wilson's algorithm; otherwise keep the forest. With K_E π-reversible on plans, the lifted kernel is $\tilde{K}(\mathcal{F}, \mathcal{F}') = K_E(P, P')/(\tau(A)\tau(B))$, and $\tilde{\pi}(\mathcal{F})\tilde{K}(\mathcal{F}, \mathcal{F}')$ is symmetric because $\tau(P) = \tau(V_i)\tau(V_j)$ times the other districts' counts, which the move does not change. Where the balance window is slack the local solve is the linear solve of Lemma 4.1, which we use in place of the QP solver; both paths are deterministic functions of the state and pair, which is all Proposition 4.3 requires.

The hybrid takes a lifted local step with probability p_{loc} and a forest step otherwise. Both kernels are $\tilde{\pi}$-reversible, so the mixture is, and its marginal on plans is π.

Proposition 4.4 (Ergodicity of the hybrid)**.** *If the forest move is irreducible on the lifted space (which holds under the standard assumptions for recombination moves; see [20, 8, 4]), then the mixture with the lifted local move at any $p_{\mathrm{loc}} \in (0, 1)$ is ergodic.*

Proof. Write $K = (1 - p_{\mathrm{loc}})K_{\mathrm{F}} + p_{\mathrm{loc}}\tilde{K}_{\mathrm{E}}$. If $K_{\mathrm{F}}^t(\mathcal{F}, \mathcal{F}') > 0$ for some t, then $K^t(\mathcal{F}, \mathcal{F}') \geq (1 - p_{\mathrm{loc}})^t\, K_{\mathrm{F}}^t(\mathcal{F}, \mathcal{F}') > 0$, so irreducibility of the forest move implies irreducibility of the mixture. The forest move rejects with positive probability at every state, so $K(\mathcal{F}, \mathcal{F}) > 0$ and the chain is aperiodic. □

Verification. On the lifted 2×3 fiber (18 labeled plans, 30 forests) we assembled the forest kernel, the lifted local kernel and their mixture analytically and checked detailed balance entrywise under both targets and both kernels: the maximum relative violation is 1.4×10^{-15}, the closed-form proposal mass (12) equals the mass obtained by enumerating every (tree, balanced edge, coin) draw to 1.8×10^{-16}, and the partition-level chains T, E and their mixture pass the same check (`forest_recom.py`, `hybrid.py`). §5.3 tests the forest move and the hybrid against enumerated targets on 4 006 plans.

4.7.4 What an exact tree move pays

Both tree moves draw the proposed split from the same law, the spanning-tree measure on balanced bipartitions of H (Chain F up to the S factor, which stays within a factor of two of one), and given the pair the split does not depend on the current one. On a merged pair the tree move is therefore an *independence sampler*, and its acceptance at equilibrium is governed by how far the target's conditional on the pair sits from that law. Two terms of (13) carry the distance. Under π_w the tree-count ratio does: a uniform spanning tree favors splits with many spanning trees, and by Procaccia and Tucker-Foltz [40] the logarithm of a tree count scales with the boundary length, so at the equilibrium of a target that is less compact than the tree measure the ratio is typically far below one. Under π_w^{tree} that ratio cancels and the target ratio $e^{-\lambda \Delta w(\partial)}$ carries the distance instead, which is large whenever the tempering makes the target more compact than an untempered tree split.

We measured the terms on Iowa over 3 000 forest proposals from a tree-partition start under π_w (Table 3). The tree-count term dominates, with mean logarithm -6 to -10 and a share of 67 to 84 percent of feasible proposals below $e^{-2.3}$ on that term alone, while the target, S and pair-count terms are mild. At the equilibrium of the uniform target the acceptance given a feasible tree is 0.17; under the tree-weighted target at the same temperature it is 0.74, because at λ equivalent to 0.1 that target is close to the untempered tree measure. The scaling experiment of §5.5 then shows what happens as the pair grows: for a compactness-tempered target, with or without tree weights, the mismatch grows in proportion to the boundary length and the acceptance decays exponentially in the grid side. The local move loses acceptance with pair size as well, because its expected move grows with the boundary (Corollary 4.1), but it has a dial for it, the timestep; the tree move has none, since its proposal is the same law whatever the target.

target	Pr[balanced]	acc. | feas.	mean of each log term				share, τ-term $< e^{-2.3}$
			target	tree count	S	bp	
uniform, $\lambda = 0.10$	0.41	0.25	+0.63	−5.93	+0.47	−0.01	0.67
perimeter, $\lambda = 0.34$	0.53	0.22	+0.65	−5.98	+0.59	−0.00	0.69
density, $\lambda = 0.72$	0.41	0.13	+1.62	−9.78	+0.74	−0.06	0.84

Table 3: The four terms of the forest move's log Hastings ratio (13) on Iowa ($k = 4$, $\varepsilon = 0.05$) under π_w, averaged over the feasible proposals of 3 000 steps from a tree-partition start, without retry. The tree-count ratio is the rejection; the retry raises Pr[tree balanced] to 0.95 at 2.5 times the wall clock and leaves accepted moves per second unchanged.

5 Experiments

We evaluate the chains of Table 1 on the 5×5 grid (full enumeration, ground-truth target), the 2×3 grid (18 states, full convergence checkable), the Iowa county dual graph (real geography, 99 counties, $k = 4$, $\varepsilon = 0.05$), and $N \times N$ grids with $k = 2$ for the scaling experiment. All hyperparameters are locked in Appendix A; raw traces, per-step balance audits, and configuration stamps are released in the project repository.

Statistical protocol. Chains are autocorrelated, so no i.i.d. statistic is reported on a raw trace. Effective sample sizes (ESS) come from Geyer's initial-positive-sequence estimator computed by FFT and are used for cut means only; distributional tests use batch means (§5.2), and distances to enumerated targets are read against parametric-bootstrap noise floors at the ESS (§5.1). Chains compared by effective

samples per second are also compared per step, and checked to sit at the same level by quarter means and by the difference of their post-burn means in standard errors. Each chain runs on its own seeded stream; baseline tables report seed 42, ten-seed means of Chains V and A (seeds 42 to 51) enter the calibration, and between-seed standard deviations of the baselines are 0.4 cuts on Iowa and 0.1 to 0.2 on the 5×5 fiber.

Balance audit. Every baseline run records at every step the maximum relative deviation of any district population from the global ideal, and all satisfy $\varepsilon = 0.05$ throughout (Iowa maxima 0.0498–0.0500); the exact chains enforce the window inside each move. This is not automatic: GerryChain's raw `recom` proposal validates only one side of each cut, and an earlier version of our local steps anchored balance to the merged-pair half, and either slip lets populations drift to four times the nominal tolerance on Iowa. We recommend per-step auditing in any harness that bypasses GerryChain's validator pipeline.

5.1 Ground truth on the enumerated fiber: what the family samples

The 5×5 partition fiber, the set of all valid plans at $k = 5$ and exact balance, has 4 006 partitions and was enumerated exhaustively, so every distributional quantity is exactly computable; at $\varepsilon = 0.05$ on unit-population cells the chains traverse precisely this fiber. In 1 000-step runs (seed 42, burn-in 100) Vanilla ReCom, tempered ReCom and Chain E have mean cut counts 17.06, 17.02 and 16.27 against the exact target mean 18.56 at $\lambda = 0.10$, and Chain E has visited only 6 states: every chain sits far below the target at this horizon, and exactness does not buy mixing speed.

The origin of the gap. The gap quantifies the compactness bias shared by the ReCom family, and on the fiber its origin can be identified. By the Matrix–Tree theorem we compute for each partition the spanning-tree-count weight $\prod_i \tau(V_i)$, the measure the RevReCom analysis identifies as the natural target of tree-based recombination [10, 4]. The tree-weighted measure has $\mathbb{E}[|\partial P|] = 16.88$ against Vanilla ReCom's ten-seed mean 16.95 ± 0.11, and the tree-weighted tempered measure $\prod_i \tau(V_i)\, e^{-\lambda|\partial P|}$ has 16.78 against Chain A's 16.71 ± 0.16; both gaps are within one between-seed standard deviation, and both quasi-targets sit about 1.8 cuts below the nominal tempered mean. A mean is a one-dimensional summary, and the long runs compare full distributions.

Long-run validation at the level of distributions. We run Chain A for 2×10^5 steps and Chain E for 10^5 steps on the fiber (seed 42, 10% burn-in) and compare each chain's empirical distribution with the nominal target, the tree-weighted measure and the tree-weighted tempered measure. Every distance is read against a parametric-bootstrap noise floor, the same distance for independent draws from the reference at the chain's effective sample size, since a raw total variation on 4 006 states is uninterpretable; and the headline statistic is the cut-count marginal, which takes five values and is estimated to three decimals. Table 4 and Figure 2 give the result.

Chain A does not sample its nominal target, and the gap is a plateau, not a drift. At 2×10^5 steps the tempered chain has visited 3 524 states, its mean is 16.73, and its cut distribution sits 220 floor standard deviations from the nominal target, at TV 0.63; Figure 2(b) shows that distance flat across five decades of run length while the distance to the tree-weighted tempered measure falls toward the floor. This is convergence to a different distribution, as the symmetric-q filter predicts. The match to the tree-weighted tempered measure is close but not exact (6.7 floor standard deviations at TV 0.022), as expected since Vanilla ReCom's stationary law is not exactly the tree measure. *Chain E samples what it states*: its cut marginal is 1.2 floor standard deviations from the nominal target and about 7 from either tree measure. Its full-state distribution converges far more slowly (403 states visited, effective sample size 28), so that test has no power here, and the fiber on which full convergence can be checked is next.

Chain	Reference measure	TV	Floor at ESS	z
A (2×10^5 steps, ESS 12 312)	nominal π	0.629	0.007	+220
	tree-weighted	0.065	0.006	+22
	tree-weighted, tempered	0.022	0.006	+6.7
E (10^5 steps, ESS 28)	nominal π	0.079	0.141	−1.2
	tree-weighted	0.493	0.125	+6.9
	tree-weighted, tempered	0.532	0.125	+7.5

Table 4: Total variation between each chain's empirical cut-count marginal and three reference measures on the enumerated 5×5 fiber, with the parametric-bootstrap noise floor at the chain's effective sample size and the excess in floor standard deviations. Chain A is far from its nominal target and close to, though distinguishable from, the tree-weighted tempered measure. Chain E is indistinguishable from the nominal target it is proved reversible for and far from the tree measures.

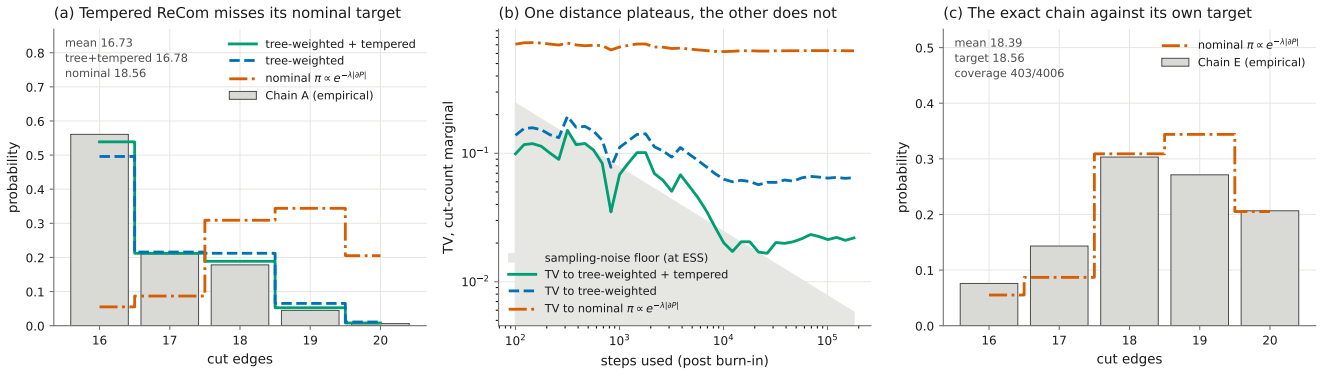


Figure 2: What the chains sample. (a) Chain A's cut-count marginal after 2×10^5 steps against the three reference measures. (b) Total variation of the running marginal to each reference as a function of run length, with the noise floor at the effective sample size shaded; the distance to the nominal target is a plateau across five decades while the distance to the tree-weighted tempered measure descends toward the floor. (c) Chain E's marginal after 10^5 steps against its own target.

5.2 Exact convergence on the 18-state fiber

Detailed balance gives stationarity, not convergence, and on 4 006 states the exact chain cannot be run to convergence in reasonable time. On the 2×3 grid with $k = 2$ and tolerance $\varepsilon = 1/3$, the fiber of labeled valid plans has 18 states, the same state space on which Remark 4.1 verifies the kernel to machine precision, and a long run can cover it. We run Chain E for 10^6 steps (seed 42, 10% burn-in, locked parameters $\alpha = 1$, $\beta = 10$, $\lambda = 0.10$). It visits all 18 states, its acceptance rate is 0.785, and its post-burn-in mean cut count is 2.75992 against the exact 2.76002.

The distributional test uses batch means: the cut count takes two values here, so its autocorrelation time (17 steps) understates that of the 18-state occupancy process (25 steps), and a chi-square test on a chain thinned by the former rejects spuriously. We split the trace into 200 blocks of 4 500 steps, estimate the occupancy covariance from the between-block scatter, and test with Hotelling's T^2 on the 17 free coordinates: $F(17, 183) = 0.87$, $p = 0.61$. Total variation to the target is 0.0110 against a floor of about 0.009, down from 0.0250 at 2×10^5 steps, the $\sqrt{n}$ rate of a converged chain. With the kernel check this is end-to-end validation of §4.5: the implemented kernel satisfies detailed balance exactly and the chain reaches its target.

5.3 The forest move and the hybrid against enumerated targets

We run the partition-level tree move (T), the forest move (F) and the hybrid (H, $p_{\text{loc}} = 0.2$, $\tau = 0.1$) from the same start for 2×10^5 steps on the 5×5 fiber (Table 5) against three enumerated targets, the uniform kernel at $\lambda = 0.1$ and at $\lambda = 1$ and a density kernel at $\lambda = 1$ built from a synthetic central city, $\rho(x, y) = 3\exp(-r^2/2)$ with $w_{uv} = \exp(-(\rho_u - \rho_v)^2)$, under which π_w favors boundaries that follow the density gradient and the tree proposal is blind to the kernel; and then the forest move and the hybrid against the tree-weighted versions of the same three targets. Each chain is tested as in §5.1, by the cut-count marginal against its noise floor and by batch-means Hotelling (cut bins of negligible target mass merged), and we report effective samples per second. The pre-registered criterion for the local move is the ratio of the hybrid's effective samples per second to the forest move's: at or above 1.5 the local move earns its place, below 1 it is a passenger.

Target	Chain	TV marg	z	Hotel. p	acc F	acc local	ESS/s	H/F
π_w, uniform, λ=0.1	F	0.006	−1.2	0.46	0.851	—	211.8	
	H	0.013	+0.2	0.43	0.851	0.660	143.8	0.68
π_w, uniform, λ=1	F	0.004	−1.3	0.80	0.979	—	338.6	
	H	0.005	−1.3	0.83	0.979	0.673	240.3	0.71
π_w, density, λ=1	F	0.004	−1.5	0.96	0.876	—	254.5	
	H	0.005	−1.5	0.25	0.876	0.784	168.9	0.66
π_w^{tree}, uniform, λ=0.1	F	0.009	+0.2	0.37	0.944	—	257.3	
	H	0.006	−0.8	0.78	0.942	0.675	188.8	0.73
π_w^{tree}, uniform, λ=1	F	0.003	−1.0	0.64	0.884	—	265.9	
	H	0.005	−0.4	0.41	0.886	0.683	182.3	0.69
π_w^{tree}, density, λ=1	F	0.007	+0.1	0.46	0.906	—	259.3	
	H	0.003	−1.3	0.70	0.906	0.789	168.6	0.65

Table 5: The forest move and the hybrid on the enumerated 5×5 fiber, 2×10^5 steps each, seed 42. "TV marg" is total variation of the cut-count marginal to the enumerated target and z its excess over the noise floor at the effective sample size; "Hotel. p" is the batch-means test of §5.2; ESS is of the weighted-cut trace.

Three findings. First, every chain samples its target: all rows sit within 1.5 floor standard deviations

of the enumerated marginal with Hotelling p between 0.25 and 0.96, under both target families and both kernels, and the same holds for Chain T (within 1.8, $p \geq 0.21$). Second, the forest move is a different baseline from the partition-level tree move: it accepts 0.85 to 0.98 against 0.07 and delivers 4 to 7.5 times the effective samples per second on this fiber. Third, on this fiber the local move is a passenger under every target, with ratios 0.65 to 0.73. That is what §4.6 predicts under exact balance, where the local move's first-order channel is closed, and it is also what §4.7.4 predicts for pairs of ten cells, where the mismatch between any of these targets and the tree measure is small. The grid tells us the construction is right and tells us nothing yet about where the local move matters; Iowa and the scaling experiment do.

5.4 The kill test on Iowa

On Iowa ($k = 4$, $\varepsilon = 0.05$) there is no ground truth, so the comparison is mobility at equilibrium. We use the paper's three kernels with λ scaled so that $\lambda\, w(\partial P)$ at the initial plan equals the uniform case's $0.1\,|\partial P|$, and run the forest move alone and the hybrid at $p_{\text{loc}} \in \{0.1, 0.3\}$, $\tau = 0.1$. The criterion was fixed before the long runs were started: under π_w, with $p_{\text{loc}} = 0.1$, the hybrid passes if its effective samples per second are at least 1.5 times the forest move's on both the perimeter and the density target on both seeds, with the two chains at the same level. The $p_{\text{loc}} = 0.3$ runs were added after the short runs favored that setting and are reported as a secondary finding.

Short runs, direction and timing. In 2×10^5-step runs under π_w the forest move accepts 0.16 to 0.17 and the partition-level tree move 0.006 to 0.012, yet the two deliver similar effective samples per second (0.16 to 0.24 against 0.15 to 0.40), because a forest step costs six times more (every feasible draw pays the determinants and the crossing-edge sums) while the partition-level move pays them only on the one percent of draws that balance. The hybrid beats the forest move on every kernel, by 1.4 to 1.8 at $p_{\text{loc}} = 0.1$ and 2.4 to 4.4 at 0.3, but the forest move's effective sample size is only 36 to 56 in these runs. Under π_w^{tree} the picture inverts: the forest move accepts 0.66 to 0.74, its effective sample size is about 10 000 per 2×10^5 steps, two hundred times its value under π_w at the same step count, and the hybrid is a passenger at 0.86 to 0.93. The per-step cost ratio of hybrid to forest move in these single-process runs is 1.00 at $p_{\text{loc}} = 0.1$ and 0.9 to 1.1 at 0.3.

Long runs, the verdict. Table 6 gives 2×10^6 steps under π_w on two seeds, and 5×10^5 steps under π_w^{tree}, where the forest move's autocorrelation time is a hundred times shorter. Under π_w the pre-registered criterion passes in all four cells: ratios 1.65 and 2.29 under the perimeter target, 1.95 and 2.03 under density, with the two chains' means within 1.5 standard errors in every cell and quarter means on the same level. The forest move's effective sample sizes are 247 to 585, which puts roughly 25 percent on each ratio, so the honest statement is "1.6 to 2.3, about two". At $p_{\text{loc}} = 0.1$ the two chains cost the same per step on this implementation, so the per-step ratios equal the per-second ones. At $p_{\text{loc}} = 0.3$ the ratios are 2.3 to 4.4; one level check there is borderline ($z = 2.0$, density, seed 43). Under π_w^{tree} the local move is a passenger with ratios 0.81 to 0.90 at effective sample sizes of 20 000 to 27 000, so those ratios are tight; per step the hybrid is slightly worse than the forest move there, so a local step contributes less decorrelation than the forest step it displaces before any cost is counted.

Reading. Against the exact global move that practice runs, the diffusion move roughly doubles effective samples per second for geography-weighted compactness targets on Iowa and adds nothing on the tree-weighted tempered measure at the same temperature. §4.7.4 says why: at λ equivalent to 0.1 the tree-weighted target is close to the untempered tree measure the forest move proposes from, so there is nothing for a local move to add, while π_w is far from it through the tree-count term. The next subsection shows that the second case is the general one.

Target	Seed	ESS		ESS/s		ratio	z	quarter means	
		F	H	F	H			F	H
π_w, 2×10^6 steps, H at $p_{\text{loc}} = 0.1$									
perimeter	42	421	696	0.123	0.202	**1.65**	+0.2	28.2 27.8 28.0 28.4	28.2 28.0 28.3 28.1
perimeter	43	247	572	0.072	0.165	**2.29**	+0.2	28.6 28.2 28.7 27.9	28.3 28.6 28.5 28.2
density	42	486	925	0.177	0.345	**1.95**	−0.3	18.0 18.0 18.3 17.9	17.9 18.0 18.2 17.9
density	43	585	1 167	0.214	0.434	**2.03**	+1.5	17.9 17.8 17.7 18.2	18.0 18.0 18.0 18.1
π_w, 2×10^6 steps, H at $p_{\text{loc}} = 0.3$									
perimeter	42	421	1 116	0.123	0.324	2.65	−0.0		28.2 28.1 28.1 28.1
perimeter	43	247	1 101	0.072	0.318	4.43	−0.9		28.2 28.3 28.2 28.1
density	42	486	1 395	0.177	0.559	3.16	+0.8		18.1 18.2 18.1 18.1
density	43	585	1 244	0.214	0.497	2.33	+2.0		18.0 18.1 18.0 18.2
π_w^{tree}, 5×10^5 steps, H at $p_{\text{loc}} = 0.1$									
perimeter	42	27 019	24 558	43.9	37.4	0.85	−0.6	13.05 13.05 13.05 13.02	13.03 13.04 13.02 13.05
perimeter	43	27 113	23 604	44.1	35.8	0.81	+0.3	13.01 13.02 13.05 13.07	13.03 13.04 13.02 13.07
density	42	21 642	19 615	32.3	29.1	0.90	−1.1	7.30 7.30 7.29 7.32	7.31 7.28 7.27 7.29
density	43	22 750	19 647	33.9	29.2	0.86	−1.0	7.30 7.30 7.27 7.30	7.27 7.29 7.28 7.29

Table 6: The Iowa kill test, long runs ($k = 4$, $\varepsilon = 0.05$, $\tau = 0.1$, 10% burn-in). z is the difference of the two chains' post-burn means in standard errors computed from their own effective sample sizes. Acceptance rates: forest 0.16 (perimeter) and 0.18 (density) under π_w, 0.73 and 0.66 under π_w^{tree}; local 0.53 and 0.67 under π_w, 0.72 and 0.84 under π_w^{tree}. All ten π_w chains ran concurrently on one machine; cost-corrected ratios (effective samples per step times the single-process cost ratio of the short runs) are 1.63, 2.29, 1.99 and 2.08 at $p_{\text{loc}} = 0.1$.

5.5 The scaling experiment

We run the forest move alone and the hybrid at $p_{\text{loc}} = 0.3$ on $N \times N$ grids with $k = 2$, so the merged pair is the whole graph and there is no pair-selection noise, unit populations, tolerance 0.10 (fixed in relative terms, so single flips are feasible at every N), uniform kernel, for $N \in \{6, 8, 10, 12, 16, 20\}$ and four targets: π_w at $\lambda = 1$ and 2 and π_w^{tree} at $\lambda = 0.1$ and 1. Each chain runs 5×10^4 steps, doubled for $N \geq 16$, from a vertical-split start. The forest move's acceptance and the two terms of its Hastings ratio are read from the hybrid's trace, where every forest proposal is made from a target-distributed state even at sizes where the forest move alone does not equilibrate within the budget.

Three findings. First, the exact tree move freezes as the pair grows, and the two tempered targets freeze it at the same rate. Under π_w at $\lambda = 2$ and under π_w^{tree} at $\lambda = 1$ the forest move's acceptance falls on a straight line in log scale, about $e^{-0.24N}$, from 0.30 and 0.22 at $N = 6$ to 0.009 and 0.007 at $N = 20$, while the per-proposal mismatch grows from about 2 to 19 in proportion to the mean cut length (7 to 25). The two curves lie on top of each other in both panels of Figure 3, although the tree-count term is present in one Hastings ratio and absent from the other. What the move pays for is the distance of the target from the untempered tree measure, not the tree-count term as such. Under the two targets near that measure the acceptance drifts from 0.8 to 0.5 over the same range and the mismatch stays below one.

Second, the local move loses acceptance with pair size too, and has a dial for it. At $\tau = 0.1$ its acceptance falls from 0.39 to 0.08 between $N = 6$ and 20, because its expected move $2\tau|\partial_w|$ grows with the boundary and multi-unit draws are more often infeasible. At $\tau = 0.03$ it is 0.74 to 0.42 over the same range. The tree move has no such dial: its proposal is the same law at every temperature and every kernel, so its acceptance is set by the mismatch, which grows with the boundary.

Third, the local move's contribution grows with pair size where the tree move pays. At $\tau = 0.03$ the hybrid's effective samples per second relative to the forest move rise from below one at $N \leq 10$ to 1.5 to

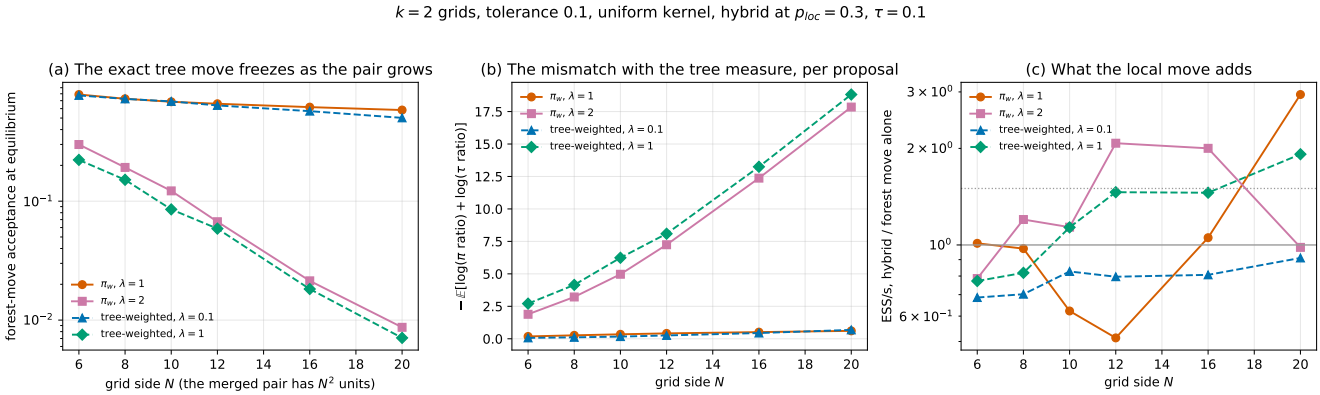


Figure 3: The exact tree move against grid side, $k = 2$, tolerance 0.1, uniform kernel, hybrid at $p_{\text{loc}} = 0.3$ and $\tau = 0.1$. (a) Forest-move acceptance at equilibrium. (b) Minus the mean of the target and tree-count log terms of (13) per feasible proposal, the per-proposal mismatch with the untempered tree measure. (c) Effective samples per second of the hybrid over the forest move alone; the $N = 20$ points of the two tempered targets are not at equilibrium (see text).

	$N = 6$	8	10	12	16	20
forest-move acceptance at equilibrium						
π_w, λ=1	0.79	0.73	0.69	0.66	0.62	0.58
π_w^{tree}, λ=0.1	0.78	0.73	0.69	0.64	0.57	0.50
π_w, λ=2	0.30	0.19	0.12	0.067	0.021	0.009
π_w^{tree}, λ=1	0.22	0.15	0.085	0.059	0.018	0.007
mean cut length under the target						
π_w, λ=2	7.2	10.0	12.6	15.1	19.8	25.1
π_w^{tree}, λ=1	6.9	9.7	12.1	15.0	20.0	25.1
per-proposal mismatch, $-\mathbb{E}[\log(\textit{target ratio}) + \log(\textit{tree-count ratio})]$						
π_w, λ=2	1.9	3.2	5.0	7.2	12.4	17.9
π_w^{tree}, λ=1	2.7	4.2	6.2	8.1	13.2	18.8
π_w, λ=1	0.2	0.3	0.3	0.4	0.5	0.6
local-move acceptance, $\tau = 0.1$ / $\tau = 0.03$						
π_w, λ=2	0.39 / 0.74	0.32 / 0.69	0.25 / 0.63	0.19 / 0.59	0.12 / 0.49	0.08 / 0.42
ESS/s hybrid over forest, $\tau = 0.03$ *(per step in parentheses)*						
π_w, λ=2	0.83 (0.78)	1.03 (0.94)	1.05 (0.91)	1.86 (1.60)	1.72 (1.46)	[3.78]
π_w^{tree}, λ=1	0.79 (0.76)	0.89 (0.80)	1.20 (1.03)	1.53 (1.31)	2.44 (2.11)	[1.32]

Table 7: The scaling experiment. Acceptance, cut length and mismatch are from the hybrid's trace at $\tau = 0.1$; the local-move acceptance and the ratios are from the $\tau = 0.03$ rerun of the two tempered targets, which the move-size law calls for once the boundary grows. Bracketed ratios are at $N = 20$, where the forest move's effective sample size is 62 and 77 and the two chains' means differ by more than two standard errors, so neither chain is at equilibrium within the budget and the ratio is not a number.

2.4 at $N = 12$ and 16 for both tempered targets, with the chains at the same level and the per-step ratios within 15 percent of the per-second ones. At $N = 20$ neither chain reaches equilibrium in 10^5 steps under a tempered target, so that size is beyond the budget of either chain and we do not quote a ratio. Under the two near-proposal targets the ratio stays below one at every size, as on Iowa under the tree-weighted target.

An exact tree move serves one family of targets well, the mildly tempered tree measure, which is what recombination samples anyway. For compactness targets away from it, in either direction, its acceptance decays exponentially in the pair size, and the diffusion move, whose reach is set by a timestep rather than by the target, is the component that keeps an exact sampler moving. The Iowa result is one point on these curves, at a pair size of about fifty units and a mismatch of about six.

5.6 Iowa: the symmetric-q filter against the exact sampler

On Iowa there is no enumeration, but the exact chains of §4.7 estimate both targets directly, and the forest move reaches the equilibrium of π_w within about a thousand steps where the partition-level tree move needed about 10^5. Under the uniform kernel at $\lambda = 0.1$ the forest move and the hybrid put the nominal target's mean cut count at 88 to 90 (half-run means 87.8 to 90.1 in the short runs of §5.4) and the tree-weighted tempered measure's at 41.0 (half-run means 41.05 and 40.93). Tempered ReCom with the symmetric-q filter, run for 500 steps from a tree-partition start (seed 42, burn-in 100), sits at 39.8 ± 4.3 cuts and Vanilla ReCom at 41.4 ± 4.8. On real geography the chain practitioners run as "targeting π" therefore samples plans with less than half the boundary length that π itself has, within one cut of the tree-weighted tempered measure, with an exact sampler on each side of the comparison.

5.7 Precinct scale

[To be written. North Carolina precinct dual graph (2 692 units, $k = 4$, $\varepsilon = 0.05$), perimeter and density targets, forest move alone and hybrid at $p_{\text{loc}} = 0.1$ with τ set by the move-size rule, 2×10^5 steps, two seeds; a timing probe first. Report forest acceptance, the two Hastings terms, quarter means, ESS per step and per second, and the ratio. Expected from §5.5: forest acceptance well below 10^{-2} under the tempered targets at pairs of about 1 300 units, hybrid reaching equilibrium where the forest move alone does not.]

6 Discussion

6.1 What each chain is for

Chain E is exact and, alone, too immobile to be a sampler; its value is as the local component. The forest move is the mobile exact component and the baseline practice runs; on its own measure it is hard to improve on, and away from it its acceptance decays with pair size. The hybrid is the practical exact sampler for kernel-weighted compactness targets, with a small local fraction on real geography and a timestep set by the move-size law. The partition-level tree move is the reference whose closed form explains why Chain A is only approximate, and Chains A and V are what practitioners run. The fidelity term is what makes the local move local: without it the move is a fresh spectral re-split that discards the structure the chain has found.

6.2 Cost, and what a compiled implementation would change

All chains here are Python. On Iowa under π_w a forest step costs about 1.1 ms, the partition-level tree move 0.18 ms, and a local step about 1.5 ms, half of which is solver overhead on the pairs where the

balance window binds. A compiled forest step would be an order of magnitude cheaper and the local step would gain less, so the per-second ratios of §5.4 would move against the hybrid in Julia or Rust; this is why every comparison also reports effective samples per step, and at $p_{\text{loc}} = 0.1$ the two coincide. Two improvements belong to a journal version: a closed-form solve for the local move when the balance constraint is active (the unconstrained solve shifted by a multiple of the population vector), which removes the solver from the local step, and a per-pair timestep $\tau = m/(2|\partial_w|)$ from Corollary 4.1 that holds the expected move at m units whatever the pair.

6.3 Limitations

No mixing-time bound: the scaling result is a straight line over six grid sizes with the independence-sampler argument of §4.7.4 as explanation, not proof. The Iowa ratios carry about 25 percent uncertainty and rest on two seeds; the direction, the level agreement and the control are solid, the second decimal is not. Chain E's irreducibility across pairs is not established, and the hybrid inherits ergodicity from the forest move. The largest real graph is 99 counties until §5.7 is supplied.

6.4 Future directions

The precinct-scale run and the downstream question it allows, whether the kernel changes an ensemble's partisan statistics. The integer form of the energy, solved as a convex mixed-integer program inside a tempered chain, is an anchored refinement tool rather than a sampler; an earlier version of this work found that it cut mean cut edges by 20 percent over tempered ReCom on Iowa and that simulated annealing over ReCom proposals reached the same best plans at a fraction of the cost, which we record so it need not be rediscovered. An informed marked-edge selection for the marked edge walk [34], with a Boltzmann draw of computable density, would be the lifted-space analogue of Chain E. The cycle walk [21] or the balanced up-down walk [1] could replace the forest move as the global component.

7 Conclusion

One Laplacian energy under global population balance gives a redistricting proposal whose optimum is a linear solve, whose expected move size is an explicit filter norm with a hard ceiling, and whose mobility is a feasibility effect with a tuning rule. Corrected exactly, it is Chain E, the first optimization-based redistricting proposal with a detailed-balance proof; lifted to spanning forests and mixed with the forest-lifted recombination move, it is a reversible sampler for kernel-weighted compactness targets that roughly doubles the forest move's effective samples per second on Iowa under perimeter and density targets and adds nothing on the tree-weighted measure. The reason is measured: an exact tree move proposes from the untempered spanning-tree measure whatever the target, its acceptance decays exponentially in the pair size under any compactness-tempered target, and the local move has a dial for the same effect where the tree move has none. On enumerated ground truth, tempered ReCom does not sample its nominal target and sits close to the spanning-tree measure; on Iowa the gap is a factor of two in boundary length. Kernel choice can be a first-class decision inside the proposal and the target, provided the induced distribution is corrected; the spanning-tree measure is the one target a tree move samples for free; and enumerated fibers read against noise floors are the severe test bed a sampler should be held to.

Code and data availability

All code, the enumerated fibers, and every artifact behind every table and figure are in the public repository

https://github.com/kirtisoglu/Redistricting_Diffusion_Model
under `diffusion_model/recom/`, with the scripts and command lines listed in Appendix B. An archived release with a persistent identifier will be deposited on acceptance.

A Algorithm details and hyperparameters

A.1 Common setup

Global two-sided balance window (4) on every move, audited per step; the bare GerryChain `recom` proposal is wrapped with a whole-plan balance check since its cut search validates only one side. Initial plans from `recursive_tree_part`, redrawn until all districts satisfy the global tolerance; a vertical split for the $k = 2$ grids. Step counts: baselines 1 000 (5×5) and 500 (Iowa) with burn-in 100; all other runs 10% burn-in, with counts as stated in the text. Seeds: base 42 with a per-chain offset, 43 for the second Iowa long run.

A.2 Chain-specific parameters

Table 8 lists every locked value.

Chain	Parameter	Value
A	λ (target temperature)	0.10
E	α, β	1.0, 10.0 ($\tau = \alpha/\beta = 0.1$)
E	Rounding	single-shot Bernoulli($x^\star$), clip 10^{-12}
T	cut edge	uniform over all tree edges; fair-coin labels
F	cut edge	uniform over balanced tree edges; fair-coin labels; retry up to 20 trees
F	pair	uniform over boundary pairs
H	p_{loc}	0.1 (Iowa long runs), 0.1 and 0.3 (Iowa short runs), 0.2 (5×5), 0.3 (grids)
H	τ	0.1; 0.03 in the rerun of §5.5
Iowa λ	uniform / perimeter / density	0.100 / 0.343 / 0.718 (equal tempering at the start plan)

Table 8: Locked-in hyperparameters for the chains of Table 1.

A.3 Compute environment

Python 3.12 on macOS and 3.13 on Linux; Gurobi 13.0 (academic license) for the QP solves the linear-solve path does not cover; GerryChain 0.2.22; NetworkX 3.x. One single-threaded process per chain; the ten Iowa long runs and the scaling chains ran concurrently on an 11-core machine, each forest chain at the same time as its hybrid partner so that both members of a ratio see the same load, and all ratios are also reported per step.

B Reproducibility checklist

All code is under `diffusion_model/recom/` in the repository. Modules: `problems.py` (graphs, balanced initial plans), `chains.py` (runners, balance-validated ReCom, per-step audit), `qp_model.py` (QP and QP-MH steps), `hybrid.py` (Chain T, Chain E with the linear-solve path, their analytic check), `forest_recom.py` (the forest move, the lifted local move, both targets, the analytic check on the lifted 2×3 fiber), `kill_stats.py` (ESS, noise floors, batch-means Hotelling). Scripts, each writing a stamped `.npz` to `iowa_hybrid_test_out/` beside its console log: `iowa_target_matched.py` (Chains V, A, E), `long_run_5x5.py`, `matrix_tree_targets.py`, `calibration_distance.py` and `fig_calibration.py`

(§5.1); `db_check_2x3.py`, `long_run_2x3.py`, `gof_2x3.py` (§4.5, §5.2); `spectral_move_size.py`, `acceptance_sweep.py`, `mobility_law.py` (§4.3, §4.6); `forest_validate_5x5.py` (Table 5); `forest_iowa.py` and `forest_verdict.py` (§5.4); `forest_scaling.py` with `forest_scaling_parallel.py` (§5.5); `hybrid_validate_5x5.py`, `hybrid_iowa.py`, `hybrid_iowa_long.py` (Chain T's rows). Command lines are in each script's docstring. The 5×5 runs and the Iowa runs under the uniform and density kernels reproduce bit for bit from their seeds; the Iowa runs under the perimeter kernel do not (a repeat matched for 21 217 and 185 390 steps and then diverged, which we attribute to floating-point sensitivity of the QP fallback under that kernel's wide weight range), and every perimeter-target conclusion rests on long runs and two seeds. [Precinct scale: data file, commands and artifacts of §5.7.]

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